# Md. Abdur Rahman Shibly

PhD Student, Computer & Information Science & Engineering, University of Florida

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## Education

**University of Florida** — *August 2026 – Present*
***Doctor of Philosophy**, Computer & Information Science & Engineering* — Gainesville, FL, USA
**Advisor:** Prof. Sumit Kumar Jha
**Research Area:** Verifiable and explainable reasoning in large language models

**Bangladesh University of Engineering and Technology (BUET)** — *March 2018 – May 2023*
***Bachelor of Science***
**BSc Thesis:**
Building a Semantic Search Engine using an Ontology: A Case Study on Social Media Influencers.

## Research Interests

- Explainable AI
- AI Reasoning
- Hallucination Reduction
- Neurosymbolic AI
- Large Language Models
- RAG
- NLP
- AI Security
- Ontology and Knowledge Graph
- Conversational AI
- HCI

## Research Experience

**Online Facility Assignment with Nonuniform Capacities** — *April 2025 – Present*
*Tools: Python, Pandas, NumPy, SciPy, Matplotlib*
Investigated the online facility assignment problem with nonuniform facility capacities in metric spaces. Established a reduction from the general setting to the unit-capacity case, simplifying the analysis of greedy algorithms. Derived tighter competitive ratio bounds for algorithms on Euclidean planes and unweighted graphs, extending prior results on uniform capacities. Manuscript under review.
Supervisors: Dr. Reyan Ahmed, Dr. Md. Saidur Rahman

**Semantic Search Engine using an Ontology** — *January 2023 – 2025*
*Tools: Protege, Neo4j, Django, Python, NumPy, Pandas*
Developed a novel ontology for social media influencers to categorize them by audience location, interaction rates, and content types. Later work refined assumptions from the original study by fine-tuning LLaMA2 and applying prompt engineering techniques.
Supervisor: Dr. Muhammad Masroor Ali

## Publications

### Conference Papers

1. Shibly, M. A. R., et al., "Leveraging the Domain Adaptation of Retrieval-Augmented Generation Models for Question Answering," in *Proceedings of the 22nd International Conference on Information Technology – New Generations (ITNG 2025)*, Springer, 2025. DOI

## Test Scores

**IELTS** (August 2025): Overall Band **8.0** (Reading: 9, Listening: 8, Speaking: 8, Writing: 6.5)

## Experience

**Graduate Teaching Assistant** — *August 2026 – Present*
University of Florida — Gainesville, FL, USA

- Teaching assistant in the Department of Computer & Information Science & Engineering alongside doctoral research

**Software Engineer** — *March 2025 – July 2026*
Bengal Mobile QA Solutions — Dhaka, Bangladesh

- Developed AI-powered systems, including AI Ruler, a dynamic parser for structuring raw CDRs
- Built chatbots and personalized AI assistants using LLMs for hotel interactions and parent-focused applications
- Led development of Invoice-Factory-AI, applying prompt engineering to extract structured data from invoices
- Modernized hotel management system with microservices-based architecture for scalability and maintainability

**Assistant Software Engineer** *July 2023 – February 2025*
Computer Network Systems Dhaka, Bangladesh

- Co-led government insurance project BISDP-BI, overseeing data analysis and ML model development for intelligent decision-making and automation
- Collaborated on AI-driven projects including Aptitudo and Zeuxis.AI, contributing to model development and data analysis
- Worked with cross-functional teams to design, test, and integrate AI/ML components into production-level solutions

**Consultant** *December 2023 – July 2024*
Bengal Mobile QA Solutions Dhaka, Bangladesh

- Developed AI assistant for hotel industry using LLMs, focusing on conversational AI solutions tailored for hotel-specific tasks
- Applied expertise in LLM-based systems for creating specialized industry applications

## Projects

**BISDP-BI** *August 2023 – August 2024*
*Government Insurance Project*
Developed comprehensive government insurance project focusing on data analysis and ML model development for intelligent decision-making regarding fraudulent claims and monetary predictions. Co-led the machine learning team throughout the entire project lifecycle.[Demo Video]
**Python, Pandas, NumPy, Scikit-learn, Machine Learning Models**

**Invoice-Factory-AI** *May 2025 – July 2025*
*AI-Powered Invoice Processing*
Developed AI system using LLM and prompt engineering to parse hotel career invoices of various formats, extracting structured data automatically. Integrated Google Document AI for OCR capabilities.
**Django, LLM, Prompt Engineering, Google Document AI**

**Multi-Agent Conversational AI System** *August 2025 – Present*
*Multi-Agent AI System*
Building comprehensive AI assistant for hotel guests, enabling coordinated task handling, service assistance, and reliable information retrieval through agent orchestration and tool-integrated workflows.
**Langgraph, Langchain, Langfuse, Multi-agent Systems**

**Zeuxis.AI** *July 2024 – December 2024*
*Multimedia Generation Platform*
Developed comprehensive platform enabling users to generate images and videos from text or image inputs, including face swapping capabilities in a unified interface. [Demo Video]
**Java, Spring Boot, FAL AI, Vision Models**

**Aptitudo** *May 2024 – August 2024*
*AI Interview Assistant*
Created AI-powered interview preparation system generating diverse question types including MCQ, coding, and written questions, with automated scoring capabilities. [Demo Video]
**Django, LLM, Prompt Engineering**

## References

**Dr. Sumit Kumar Jha**
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*Relationship: Doctoral Advisor*

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*Relationship: BSc Thesis Advisor*

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*Relationship: Research Supervisor*